

Q. What is Wiringpi? I am a PHP programmer. Can I use it?

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Through e-mail

A. Arduino is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. The microcontroller on the board is programmed using the Arduino programming language based

on 'Wiring' and the Arduino development environment. 'Wiring' is the core software that is used to control various ports of Arduino development board. Wiringpi is a similar package that works on Raspberry Pi. Commands are the same as for Arduino, and various sensors, when connected to Rasp, communicate with the Raspberry pi like Arduino does. Both Wiring and

update using below command:

```
sudo apt-get update
3. You need to install php5-dev before installing WiringPi-PHP. The command for installation is:
```

```
sudo apt-get install php5-dev
4. Now install git to install WiringPi-PHP using command:
```

```
sudo apt-get install git
5. Clone the WiringPi-PHP repository using:
```

```
git clone https://github.com/WiringPi/WiringPi-PHP.git
```

6. Use 'cd' command and go to the downloaded directory:

```
cd WiringPi-PHP
```

7. Now write 'git submodule update -init' in the terminal and press 'Enter'

8. Use the following two commands to build and install WiringPi-PHP:

```
sudo ./build.sh
sudo ./install.sh
```

9. Copy 'wiringpi.php' to '/var/www/' location using command:

```
cp wiringpi.php /var/www/
```

Now Wiringpi is installed on your Raspberry Pi and also copied to the /var/www directory.

To get it inside PHP, you need to load it in the '.ini' file using following steps:

1. Open a 'wiringpi.ini' file from '/etc/php5/conf.d/' in nano text editor using command:

```
sudo nano /etc/php5/conf.d/wiringpi.ini
```

2. Write the following lines in it:

```
extension=wiringpi.so
wiringpi.pinnamtype=PINS
```

Save it using 'ctrl+s' and exit using 'ctrl+x'

3. Restart apache server using:

```
/etc/init.d/apache2 restart
```

4. Run the following command to check whether the Wiringpi module is loaded or not:

```
php -m | grep wiringpi
```

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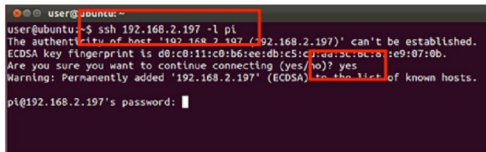


Fig. 1: Remote access using Linux system (SSH should be enabled on Raspberry Pi)

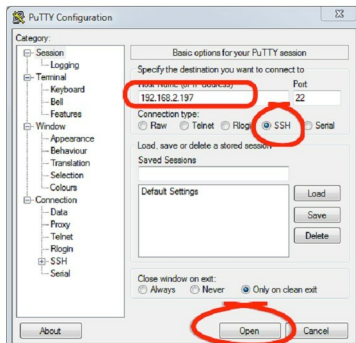


Fig. 2: PuTTY can be used to remotely access Raspberry Pi from a Windows system

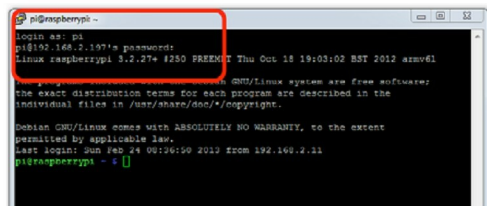


Fig. 3: Remote terminal session on Windows system

Wiringpi are open source software, developed by Gordon Henderson.

You can use the PHP version of Wiringpi and make applications with PHP programming language.

Install WiringPi-PHP by following below-mentioned steps:

1. Open LxTerminal on Raspberry Pi or open a terminal session using SSH as shown in

Figs 1, 2 and 3

2. First, do an